

DeepTrace

Multi-Agent AI Research System

Project Documentation

An AI-powered research workflow that searches the web, collects useful information, writes a report, reviews it, and improves it when needed.

1. Introduction

DeepTrace is a practical AI research system built to make the research process easier. Instead of asking one AI model to do everything in a single response, DeepTrace breaks the work into separate stages. One part searches for information, another collects useful content from those sources, another writes the report, and a separate AI reviews the result.

The idea is simple: good research is not only about writing an answer. It also involves finding information, reading it, putting it together clearly, and checking whether the final result is good enough. DeepTrace tries to automate this complete flow in one application.

2. Problem Statement

Researching a topic manually can take a lot of time. We search for articles, open different websites, read through them, collect important points, write a summary, and then go back and check whether something important was missed.

A normal chatbot can generate a report quickly, but that does not automatically mean it has followed a proper research process. There is also no clear separation between finding information, writing the report, and checking the quality of that report.

DeepTrace addresses this by turning research into a step-by-step workflow where different agents have different responsibilities.

3. Objective

The main goal of DeepTrace is to build a working example of how multiple AI agents can cooperate to complete a real task. The system is designed to:

- take a research topic from the user
- search the web for relevant information
- collect readable content from useful sources
- turn that information into a structured report
- review the generated report using a separate AI model
- rewrite the report when the review identifies problems
- stop after a limited number of attempts instead of looping forever
- show the complete process through a simple web interface

4. Target Audience

DeepTrace can be useful for students, developers, researchers, and anyone who wants a quick starting point for understanding a topic. It is also useful for people learning Generative AI and agent-based systems, because the project shows how individual AI tasks can be connected into a larger workflow.

5. Solution

The solution is a multi-agent research pipeline. The user only needs to provide a topic. DeepTrace then handles the rest of the workflow automatically.

Search → Research → Write → Critique → Improve (if needed) → Final Report

The search and research stages are performed once. If the report needs improvement, only the Writer and Critic work together again. This keeps the revision process focused instead of repeating the entire research process.

6. Methodology

The workflow is managed using LangGraph. Each major step is represented as a node, and the current research state is passed from one step to the next.

- The user enters a research topic.
- The Search Agent finds relevant sources using Tavily.
- The Research Agent fetches and extracts useful webpage content.
- The Writer Agent creates the first draft using Mistral.
- The Critic Agent reviews the draft using Gemini.
- A Router decides whether the report is finished or needs another writing attempt.
- The final report is displayed in the Streamlit application.

7. How the Agents Work

7.1 Search Agent

The Search Agent uses Tavily to find relevant web results. It collects useful search information and URLs that can be used in the next stage.

7.2 Research / Scraper Agent

The Research Agent opens the collected URLs and extracts readable text from the pages. Requests is used to fetch the pages and BeautifulSoup is used to parse the HTML. This gives the writing stage actual source material to work with.

7.3 Writer Agent

The Writer Agent takes the collected information and turns it into a proper research report. Mistral Small is used for this stage. The aim is to make the information understandable and organized rather than simply copying pieces of scraped content.

7.4 Critic Agent

Once the report is written, a separate Critic Agent reviews it. Gemini 2.5 Flash is used for this stage. The Critic provides feedback and an approval or rejection decision.

7.5 Revision Loop

If the Critic rejects the report, the Writer gets another chance to improve it using the feedback. This creates a simple generate → review → improve cycle. The workflow allows a maximum of three attempts, so it cannot keep revising indefinitely.

8. Architecture

Stage	Responsibility
User Query	Provides the research topic
Search Agent	Finds relevant web sources using Tavily
Research Agent	Fetches and extracts webpage content
Writer Agent	Creates the report using Mistral
Critic Agent	Reviews the report using Gemini
Router	Decides whether to finish or revise the report

9. Technology Stack

Technology	Purpose
Python	Core application and agent logic
LangGraph	Workflow orchestration and conditional routing
Streamlit	Web interface and deployment
Mistral Small	Research report generation
Gemini 2.5 Flash	Report critique and feedback
Tavily	Web search
Requests	Fetching webpage content
BeautifulSoup	HTML parsing and text extraction
python-dotenv	Environment/API key management

10. User Interface

The frontend is built with Streamlit so the project can be used without a separate frontend framework. The interface shows the research progress, agent stages, generated report, sources, and research information. It also provides a way to download the final report.

A small but important usability feature is that the research action is disabled while a run is in progress. After a run finishes, the same run cannot be accidentally started again unless the user chooses to start new research.

11. Results

DeepTrace successfully completes the full research pipeline from a topic to a final report. During testing, the system was able to search for information, scrape source pages, generate a report, evaluate it, and revise it when the Critic found that another attempt was needed.

For the topic “Impact of AI on Jobs 2026”, the workflow completed in two attempts. This showed that the Critic was able to send the report back to the Writer for improvement before the process finished.

The result is a working prototype of an autonomous research assistant rather than a simple single-prompt chatbot.

12. Deployment

DeepTrace is deployed as a Streamlit web application. The application connects the frontend to the research workflow and external AI/search services through configured API keys.

Live Application: deeptrace.streamlit.app

13. Conclusion

DeepTrace is a practical demonstration of how AI agents can work together to solve a task that normally requires several manual steps. Instead of treating an LLM as a single all-purpose chatbot, the project gives different components clear jobs and connects them through a controlled workflow.

The Writer–Critic loop is the main idea behind the project. Rather than assuming the first generated report is good enough, the system gives another AI the responsibility of reviewing it and asking for improvements when necessary.

Overall, DeepTrace brings together **Generative AI, web research, multi-agent workflow orchestration, and iterative evaluation** in one working application.